

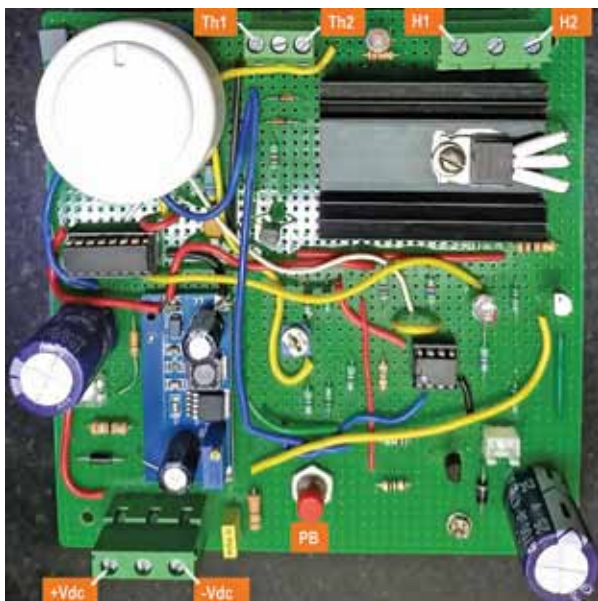


Fig. 5: Cooker PCB with assembled components

heater coil is connected to terminals H1 and H2. Similarly, connect the thermistor to terminals Th1 and Th2. When the power is switched on, MOSFET IRF1 is off, hence terminal H2 is at 48V. This voltage is applied to the base of NPN transistor T2 through potential divider formed by R13 and R14. T2 is on and it pulls down the gate of MOSFET to ground; it remains off.

To turn on the cooker, the user has to press the pushbutton S2. T2 turns off and MOSFET turns on. This is because, initially pin 3 of IC1 is low, hence PNP transistor T1 is on. Pin 3 of IC1 allows base current to flow through resistor R10 and diodes D2 and D3.

The green LED2 shows that the heater is on, connected between H1 and H2 with current-limiting resistor R3. As the heater warms up, the thermistor's resistance decreases. When it falls below a threshold ($< R15 + VR2$), COMP2 output goes low, triggering the counter IC1 to start counting. Once the user-set time elapses, the counter's output at pin 3 goes high, turning off transistor T1 and the MOSFET.

In Fig. 4, an alarm circuit is included. It comprises NPN transistor T3 with a buzzer (PZ1) connected to its collector. The base of T3 is linked to counter IC1's pin 3 via resistor R25. When pin 3 of the counter goes high, T3 is activated. Capacitor C4 is gradually charged by R26 during the cooking phase. Capacitor C4 discharges through D4, the buzzer, and T3, causing the buzzer to sound until C4 is

fully discharged.

Construction and testing

PCB fabrication. Fig. 5 displays the PCB with labelled external connections. In the upper left corner, you will find the potentiometer VR1 with a white knob, which allows users to set the counter time.

Fig. 6 displays the entire electrical system. The AC-DC converter is

linked to the 230V AC mains supply via an unshown switch S1. Its output is connected to the PCB. The heater and thermistor are connected to screw terminals, while the buzzer is linked to the PCB through a 2-pin connector.

Heater assembly. As shown in Fig. 7, the heater assembly comprises a 1800W heater coil and a thermistor. Choose a heater coil with three legs and attach three small wooden battens to these legs to minimise heat loss. Connect Teflon wires to the heater's two terminals. Mount the thermistor at the coil's centre on the leg support structure, connecting it with two Teflon wires. Exercise caution to shield all electrical joints from exposure to air; wrapping multiple layers of Teflon tape is recommended to ensure it remains secure during daily use.

Never heat the coil without a vessel placed on it. This cooker is designed for food items with water content or when water is added to the vessel. Using it for dry food items can harm the thermistor and other plastic components.

Fig. 8 shows the placement of the heater within the casserole



Fig. 6: Complete electrical system with all interconnections